Literature Status: The Critical $H^{1/2}$ Non-Autonomous Maximal-Regularity Case

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Status. **literature_already_answered**, under the uniform Kato square-root and local integrability hypotheses made precise below.

Original source. Fackler's paper [1] reviews the Sobolev time-regularity threshold for non-autonomous forms. On PDF page 2, in the introduction (parsed source line 214), it states:

> What remains open as of now is the critical case of $H^{1/2}$-regularity.

Here the problem is maximal L^2-regularity on the pivot Hilbert space for the Cauchy problem $u'(t) + A(t)u(t) = f(t)$, where $A(t)$ is induced by a time-dependent bounded coercive form.

Separate later answer. Achache and Ouhabaz [2] explicitly identify this target in their introduction. On PDF page 3 (parsed source lines 140–142), after calling fractional $H^{1/2}$-regularity the remaining problem, they write "We solve this problem in the present paper." Their Theorem 2.2, on PDF page 5 (parsed source lines 190–195), says the following. Suppose the forms satisfy the standard measurability, boundedness, and coercivity hypotheses and the uniform Kato square-root property. If

$$t \longmapsto \mathcal{A}(t) \quad \text{is piecewise in} \quad H^{1/2}\big(0, \tau; \mathcal{L}(V, V')\big)$$

and satisfies their local condition (2.1), then the problem has maximal L^2-regularity in the pivot Hilbert space for every $u_0 \in V$, with the corresponding a priori estimate.

Condition (2.1) requires that, after a sufficiently fine finite subdivision,

$$\sup_{t \in (\tau_{i-1}, \tau_i)} \int_{\tau_{i-1}}^{\tau_i} \frac{\|\mathcal{A}(t) - \mathcal{A}(s)\|^2_{\mathcal{L}(V,V')}}{|t-s|}\, ds$$

be arbitrarily small on every subinterval. The paper notes that this holds, for example, under any positive Hölder modulus and under a more general square-Dini modulus.

Identification. This is a direct later-paper answer, not a same-paper ask-and-answer passage. The later paper names the same threshold, declares that it solves the remaining problem, and proves the critical theorem. It was submitted in September 2017, after arXiv:1609.08823 (September 2016).

Scope limitations. The answer is not a theorem from bare $H^{1/2}$-membership alone: Theorem 2.2 also assumes uniform Kato and condition (2.1). This packet records the positive critical result in exactly that formulation. It does not resolve two different remarks in Fackler's source: coefficient-independence of the endpoint real trace space for divergence-form operators on L^q, and whether every uniformly Kato family admits an L^2-trace parameterization.

Novelty check. The run's cheap indexes had no existing status packet for either arXiv id. Exact-title and exact-phrase searches located arXiv:1709.04216, and its local parsed source supplied the explicit identification and theorem above. No new mathematical solution is claimed here.

References

[1] S. Fackler, *Non-Autonomous Maximal Regularity for Forms Given by Elliptic Operators of Bounded Variation*, arXiv:1609.08823; J. Differential Equations **263** (2017), 3533–3549.

[2] M. Achache and E.M. Ouhabaz, *Lions' maximal regularity problem with $H^{1/2}$-regularity in time*, arXiv:1709.04216; J. Differential Equations **266** (2019), 3654–3678.